

**WELCOME
&
THANK YOU**

<Creative  Software/>

**Spring/Spring Boot
Crash Course**

For TD Bank

MEET A FEW OF OUR TEAM MEMBERS



TANGY F.
CEO



LINA G.
PROJECT MANAGER



CHEDDAE G.
TECH ASSISTANT

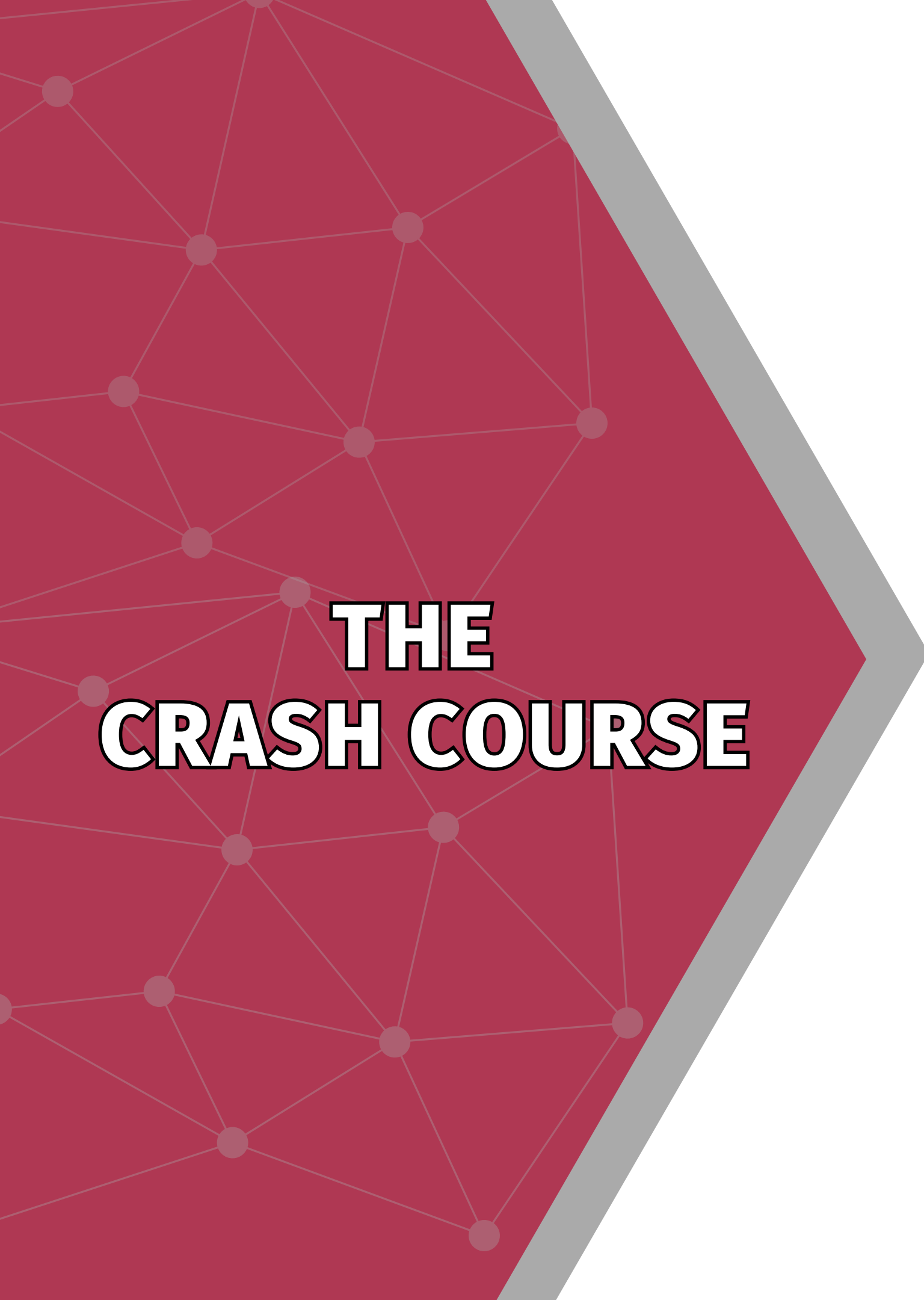


HAL M.
DEVELOPER



WILLIAM D.
DEVELOPER

- ***Founded in South Florida in 2018***
- ***Evolved from Software Development Expanded into a Technical Training firm Providing tailored technical training solutions for businesses***



THE CRASH COURSE

**Crash Course is our
bite size lighting training**

**The Duration is 1 hour a day
for 10 days**

**One Topic~
In this case it's Spring/Spring
Boot**

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**WE LISTENED TO
YOU & YOUR
COLLEAGUES**

- ① **We interviewed practitioners and your leads**
- ② **We reviewed anonymous data**
- ③ **Came up with a tailored curriculum based on real-time data**

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THE SPACE

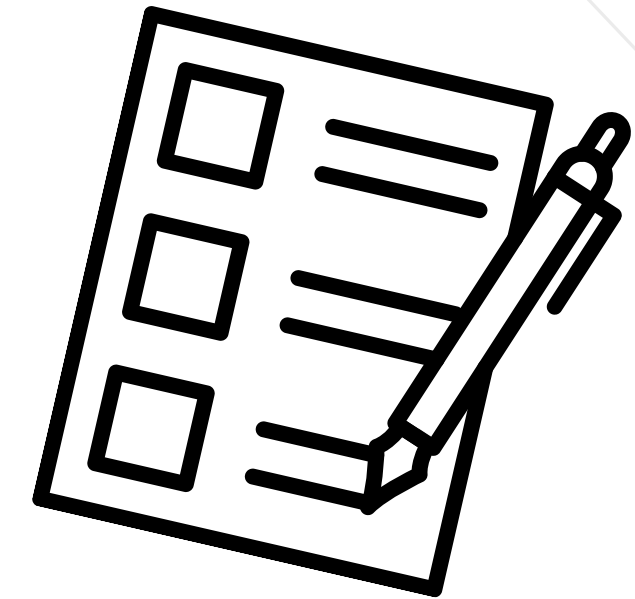
**A safe space where our goal
is to help you**

**We are here to help you
grow. If there is something
you don't understand this is
the place to ask.**

This is a 'No Judgment Zone'

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TIME SET ASIDE



Pre Assessment

MEET YOUR CRASH COURSE TEAM



TANGY F.
CEO



HAL M.
DEVELOPER



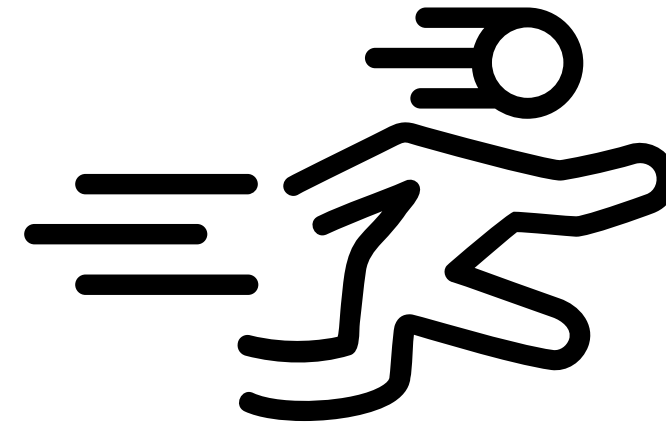
WILLIAM D.
DEVELOPER

AGENDA

TRAINING DAY 1

- 1 What we are building
- 2 Lesson 1 Dependency Injection REST API
- 3 Q.A. 10 mins

SPRING / SPRING BOOT



Rapid overview
Spring / Spring Boot



JDK 11 VS JDK 17

JDK 11

- Active Support through Sept. 2023
- Extended Support through Sept. 2026
- Getters, Constructors, toString etc. required to be created to define functionality of a class.
- Tells you when a NullPointerException has happened, but not which chained method caused it.
- collect() method is required to output a Stream to a List.

JDK 17

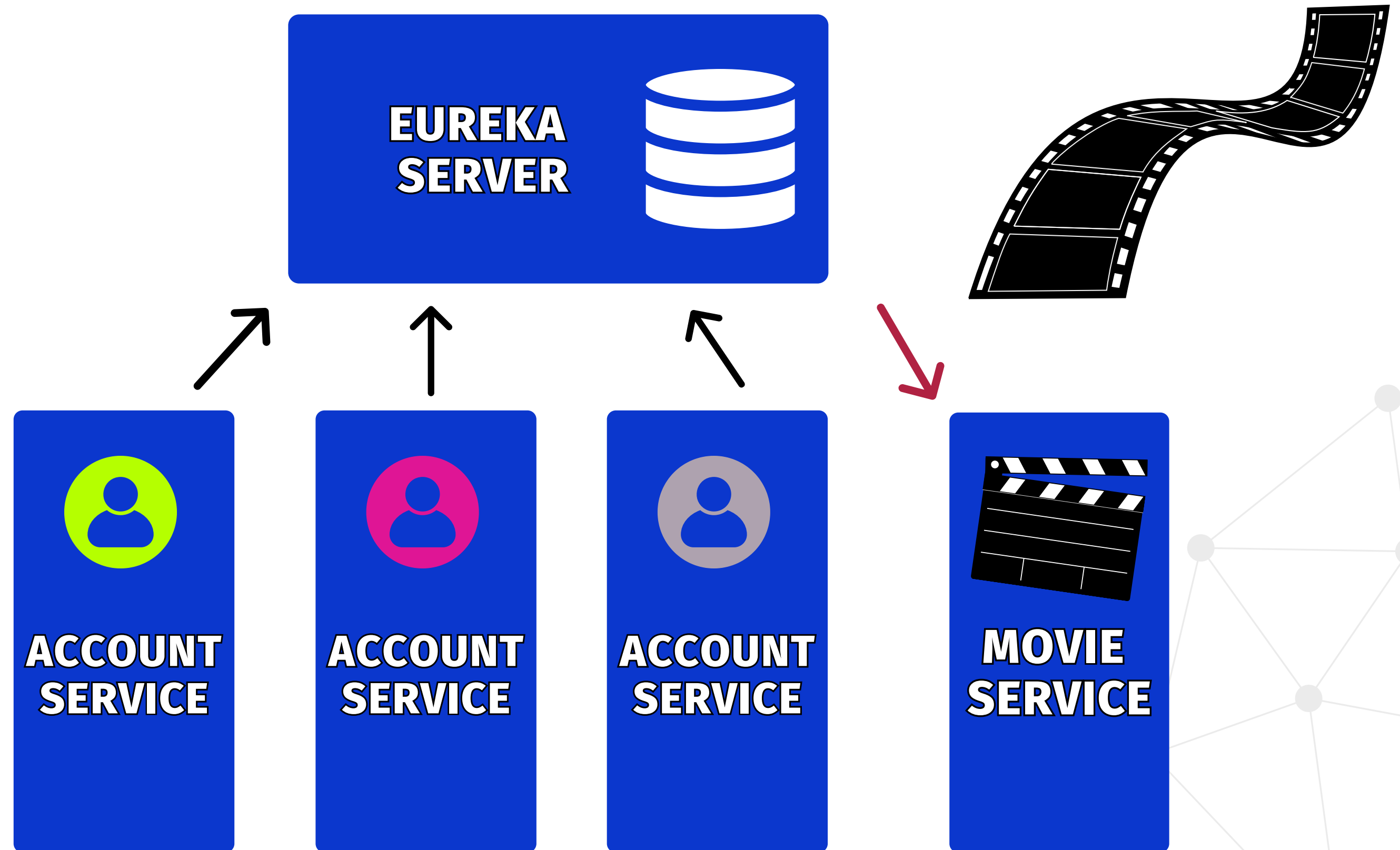
- Active Support through Sept. 2026
- Extended Support through Sept. 2029
- Record object can be created to define a simple object with variables in a single line with data methods built in.
- Shows exactly where all NullPointerExceptions occur.
- toList() method converts a Stream directly into a List.

LESSON 1

Dependency Injection REST API

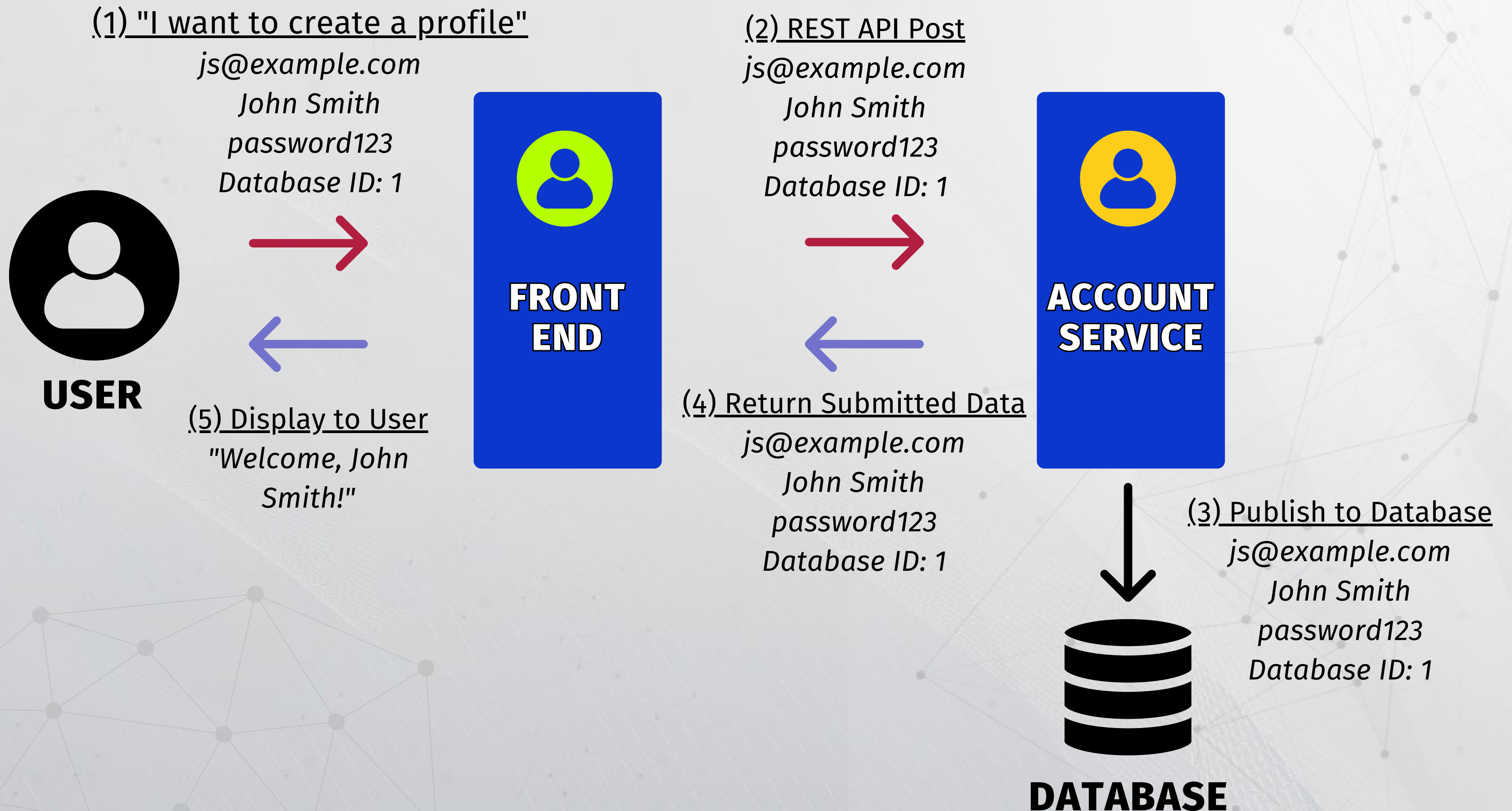
WE ARE BUILDING

**BITE SIZE
MOVIE REVIEW APP**

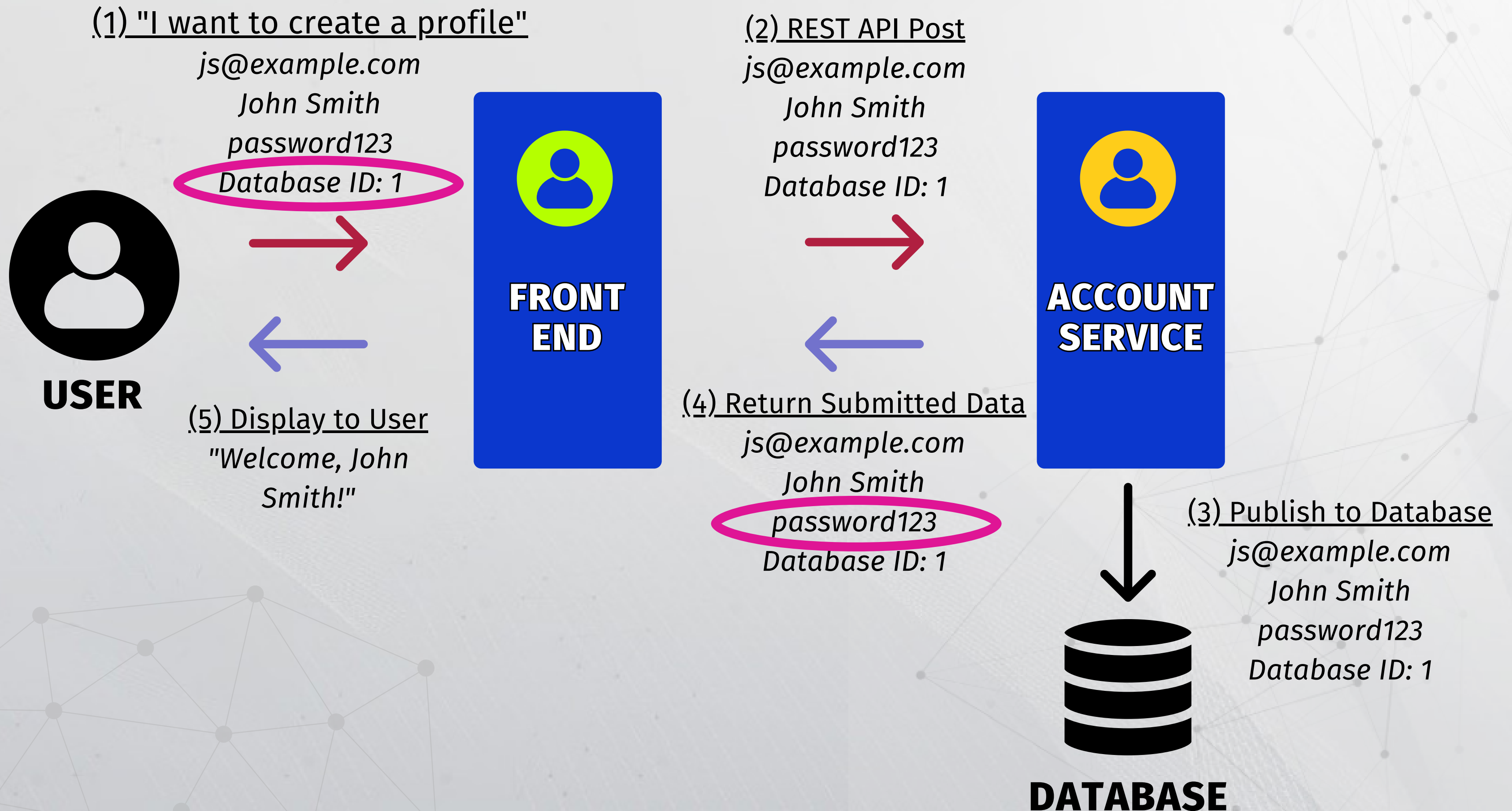


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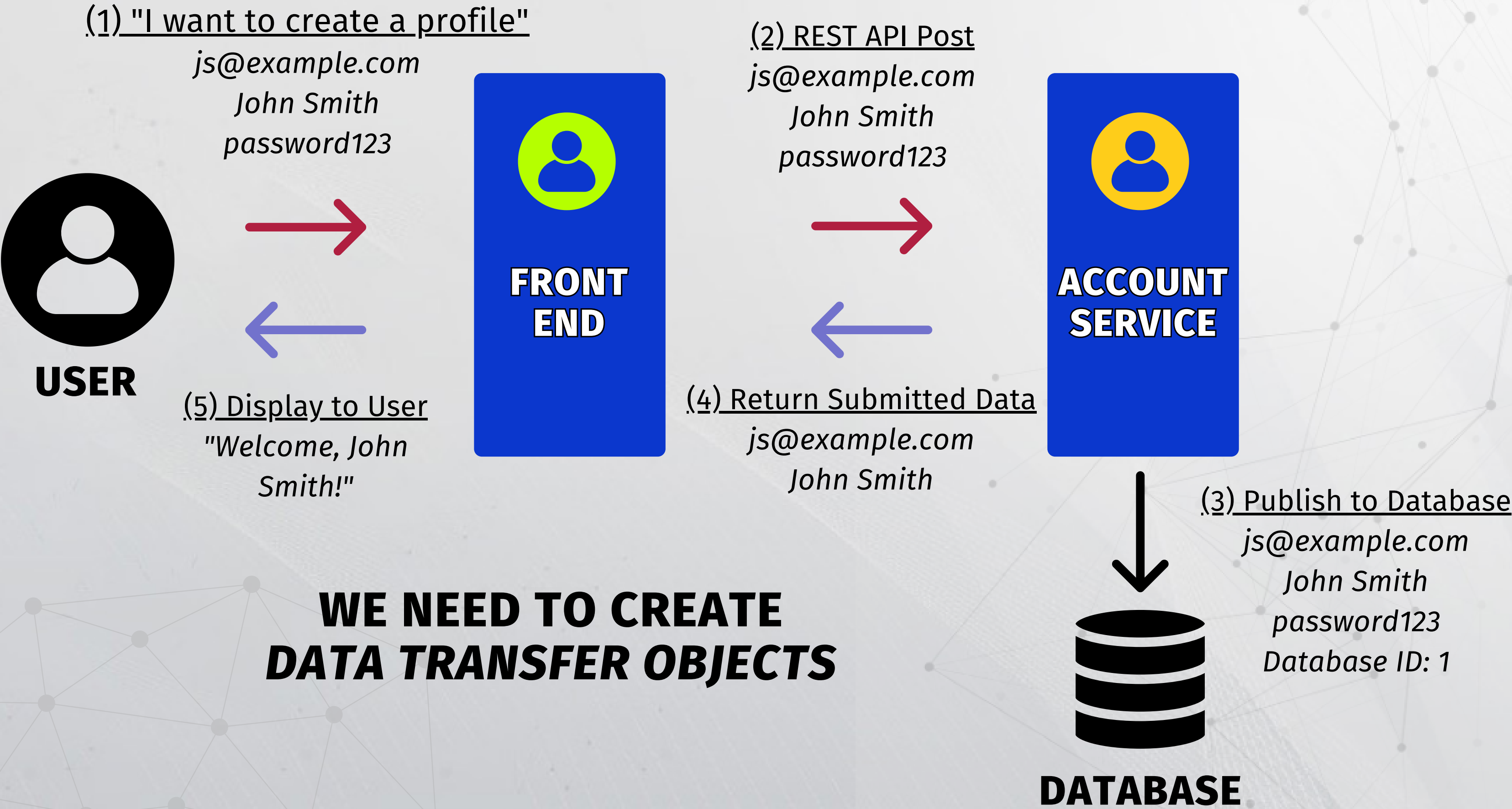
USE CASE : CREATING A PROFILE



USE CASE : CREATING A PROFILE



USE CASE : CREATING A PROFILE



TIME TO CODE #1



The below code takes an Account object as a parameter, and returns the Account object back.

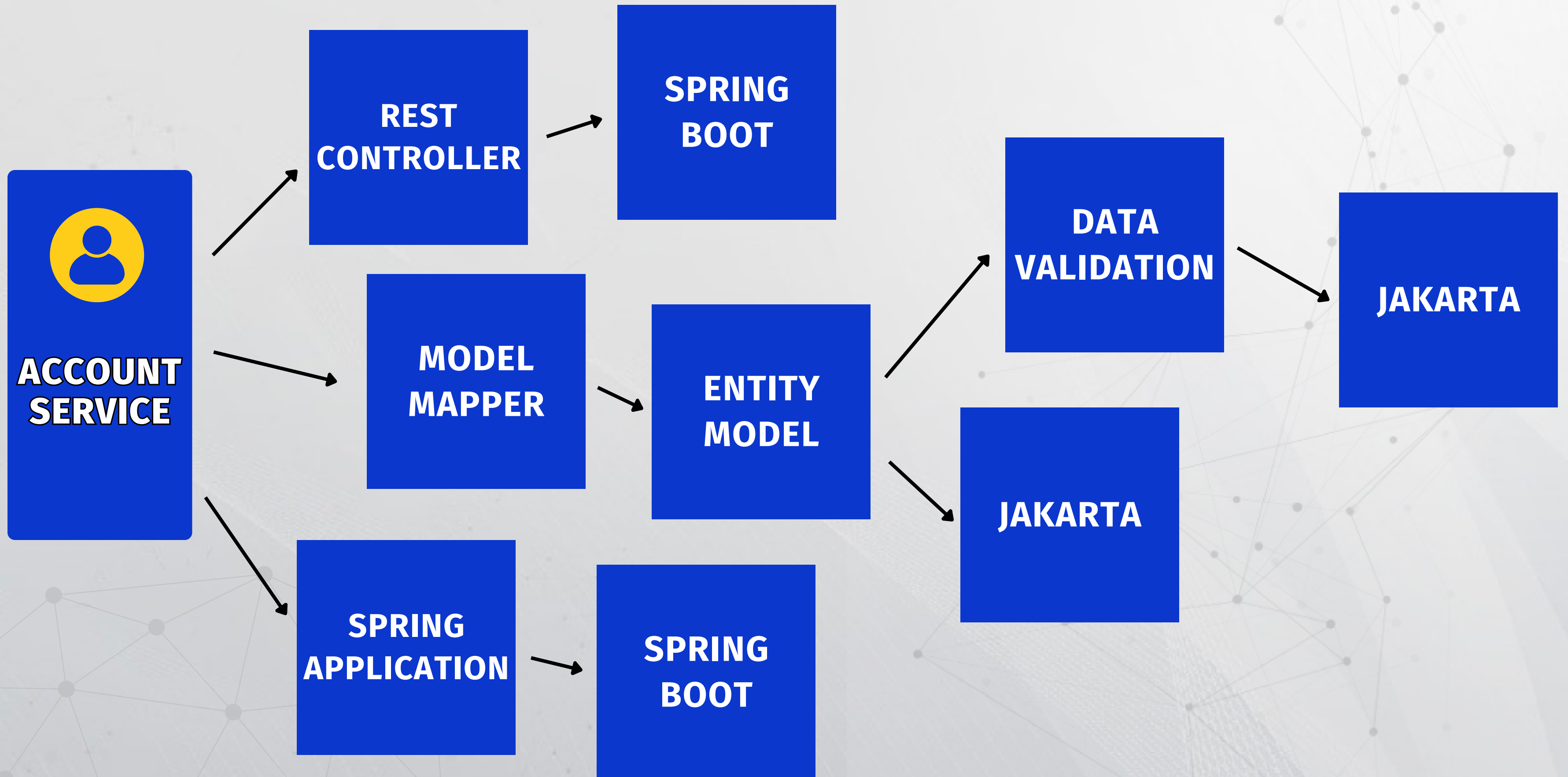
Rewrite the code so that the function takes an AccountRequestDto object as a parameter, and returns an AccountResponseDto object that has the same *name*, *address*, *email*, and *mobNo* fields as the original AccountRequestDto object.

Use proper dependency injection/inversion of control.

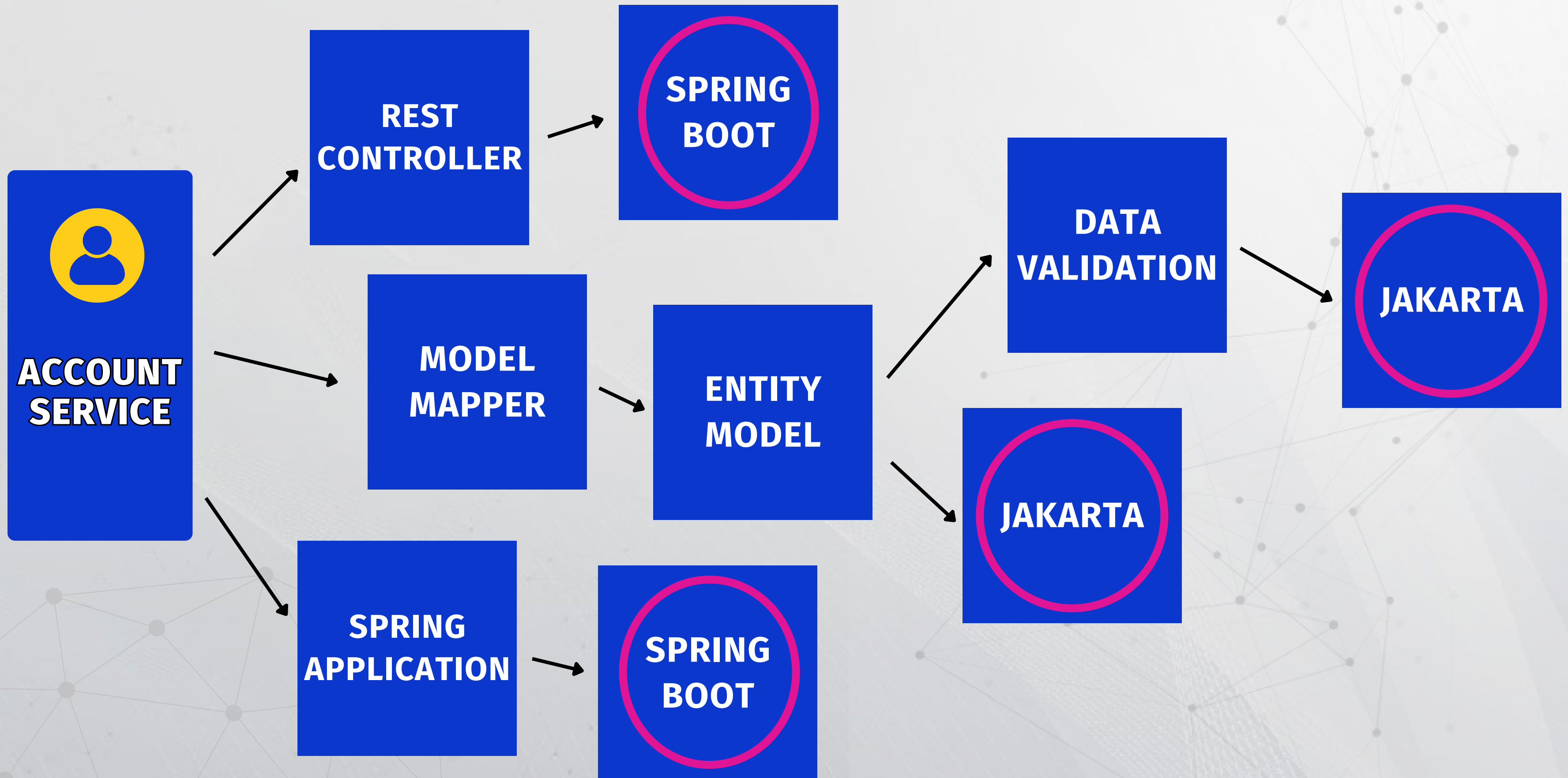
```
@Autowired
ModelMapper modelMapper;

public Account createAccount(Account account) {
    return account;
}
```

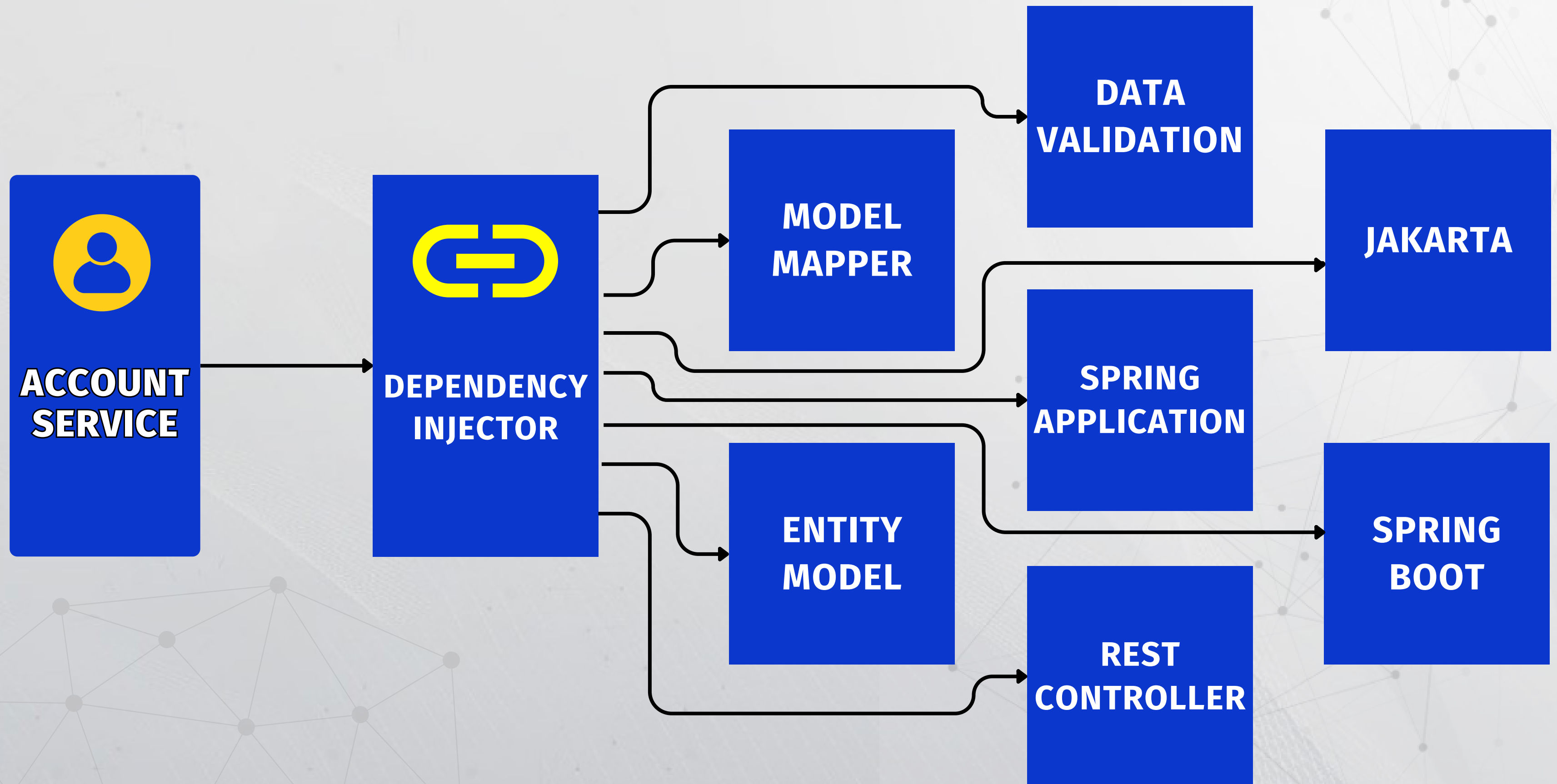
WHY DEPENDENCY INJECTION?



WHY DEPENDENCY INJECTION?



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DEPENDENCY & REST API

Modes of DI

Constructor Injection
Setter Injection
Field Injection

Entity to DTO

Modeling or Mapping the Entity data to the format that is required. It can be of any type JSON, XML.,

Scope of a bean

Singleton
Prototype
Request
Session
Global

REST API

Request Mapping from
Controller to Service

TIME TO CODE #2



The below code is our REST API endpoint for creating an account in our AccountController class. It is intended to take an AccountRequestDto as a parameter using Jakarta validation, and returns a ResponseEntity containing an AccountResponseDto as a response.

The code currently does not work as intended. What is missing from this code?

```
@PostMapping
private ResponseEntity<AccountResponseDto> createAccount(AccountRequestDto accountRequestDto) {
    return new ResponseEntity<>(accountService.createAccount(accountRequestDto), HttpStatus.OK);
}
```



THANK YOU

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We will see you tomorrow

